5 A length of conducting wire is bent into the shape of an isosceles right-angled triangle, where the two equal sides are each 0.15 m in length as shown in Fig. 5.1.

The loop is placed in a uniform magnetic field of flux density 0.40 T directed perpendicularly into the plane of the loop. The loop moves at a constant velocity of 0.30 m $s^{-1}$ in the plane of the loop to the right.

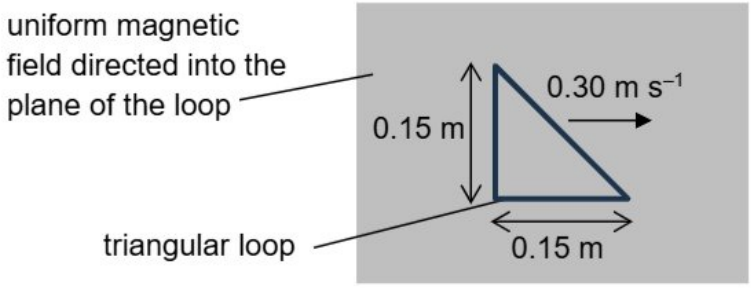



**Fig. 5.1**

**(a)** Explain why an e.m.f. is induced in the loop as it moves out of the magnetic field.

.....

.....

.....

.....

..... [2]

**(b)** The loop starts to exit the field at time $t = 0$ and it fully exits the field at time $t = T$.

On Fig. 5.2, sketch the variation of the induced e.m.f. $E$ with time $t$.

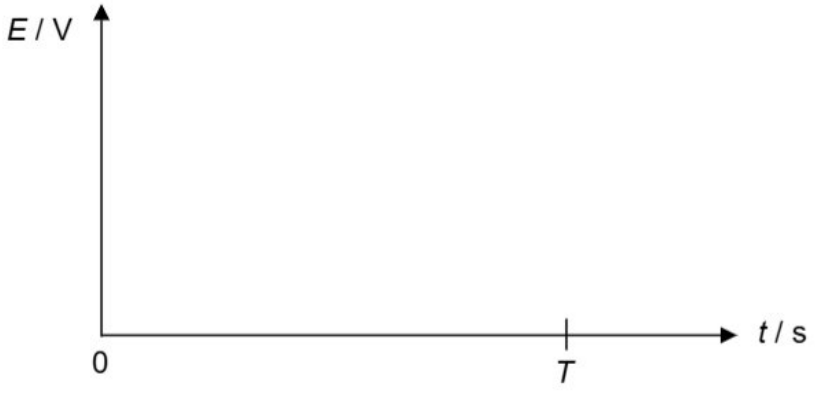



**Fig. 5.2** [1]

**(c)** Determine the average e.m.f. induced in the loop.

e.m.f. = ..... V [2]

[Total: 5]